

Amro A. Abdalla

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Education

- **Georgetown University** Aug 2023 – May 2028
Ph.D. in Computer Science Washington DC, USA
 - **Research Advisor:** Prof. Sarah Adel Bargal
 - **Research Interests:** Computer Vision, Multi-modal Learning, VLMs, World Modeling
 - **Awards:** Fritz Family Fellowship - Interdisciplinary research fellowship focused on the social impacts of technology
 - Best Teaching Award Spring 2025.
 - **GPA:** 3.9/4.0
- **The American University in Cairo – AUC** Feb 2017 – May 2022
B.Sc. in Computer Engineering Cairo, Egypt
 - **Awards:** Honors (Cum Laude), Dean's List, Full Merit-based scholarship from Al-Ghurair Foundation
 - **Minors:** Mathematics, Business Administration
 - **Major GPA:** 3.6/4.0

Selected Graduate Courses

Deep Learning for Computer Vision • Empirical Methods in NLP • Experimental AI • Deep Learning • Data Science • Big Data and Cloud Computing • Human Centered CS • NLP Seminar

Papers

1. **Abdalla A**, Shaheen I, DeGenaro D, Mallick R, Raita B, Bargal SA. (2025) GIFT: Gradient-aware Immunization of diffusion models against malicious Fine-Tuning with safe concepts retention [arXiv preprint arXiv:2507.13598, 2025](#) [*In submission*]
2. Evaluating Concept Filtering Defenses against Child Sexual Abuse Material Generation by Text-to-Image Models [*In submission*]
3. Mallick R, **Abdalla A**, Bargal SA. FaithFill: Faithful Inpainting for Object Completion Using a Single Reference Image. (2025) [arXiv preprint arXiv:2406.07865, 2025](#). [*In submission*]
4. Cui, C., **Abdalla, A.**, Wang, D., Solberg, S., Bargal, S.A. (2024). Large Language Models for Career Readiness Prediction. AIED 2024. DOI: [10.1007/978-3-031-64315-6_26](#) [*Nominated for best poster award*]

Industry Experience

- **Dell Technologies** Oct 2022 – Aug 2023
Data Scientist - R&D Team Cairo, Egypt
 - Co-inventor on two patent applications for applying ML to optimize 5G network operations.
 - Developed ML algorithms for intercell interference detection in 5G networks, improving detection accuracy.
 - Collaborated with international teams (Canada and others) to develop and integrate ML solutions into production.
 - Mentored two teams in Dell's *Envision the Future AI Competition*; one placed 1st in the 2022–2023 META region.
- **Dell Technologies** Mar 2022 – Oct 2022
Data Science Intern - R&D Team Cairo, Egypt
 - Built and evaluated ML models for 5G admission control, achieving a 12% performance improvement over fixed threshold-based methods.
 - Curated and analyzed data from Network Simulator 3 (ns-3), generating visual insights to inform model development.
 - Led the end-to-end planning and execution of Dell's annual *HackTrick* hackathon for professionals and students.
- **Siemens EDA (Mentor Graphics)** Aug 2021 – Oct 2021
Software Development Intern Cairo, Egypt
 - Developed a C-based parsing tool for Siemens' proprietary UCDB database using UCDB APIs and the `libxml` library to export structured XML and text representations.
 - Integrated the tool into Siemens' internal workflows, enabling automated analysis and downstream processing.

- The American University in Cairo**
Research Intern - Systems Lab

- Built a stack unwinding library for NileOS, a distributed microkernel OS for high-performance applications, using C, C++, and x86 assembly.
 - Designed a g++-compatible exception handling framework integrated with NileOS.

Jan 2020 – Jun 2020

Cairo, Egypt

Teaching Experience

- Georgetown University**
Graduate Teaching Assistant - COSC 3470: Deep Learning

Feb 2025 - Current
 - Georgetown University**
Graduate Teaching Assistant - COSC 3510: Intro to Databases

Aug 2024 - Dec 2024
 - The American University in Cairo**
Teaching Assistant - CSCE 4604: Practical Machine Deep Learning

Aug 2021 - Dec 2021

Projects

- PaliGemma:** Implemented a multimodal (vision-language) model from scratch using Python and PyTorch, including custom transformer architecture, image-text fusion, and end-to-end training pipeline. It combines the SigLIP vision encoder and the Gemma language model. It was introduced by Google DeepMind
- Garment Is All You Need:** Developed an inference pipeline using diffusion models and IP-adapters to generate personalized fashion model images, enabling customization through text prompts and portrait inputs.
- Custom AI Assistant with Retrieval-Augmented Generation (RAG):** Built a legal AI chatbot with Llama 2, FAISS, and LangChain, supporting file uploads, web search, and contextual queries.
- Analyzing U.S. Crime Patterns:** Curated and analyzed U.S. crime datasets using statistical methods, clustering, and topic modeling, and developed an interactive website with Plotly to showcase trends and insights.

Talks

- Dsquares Tech Meetup**
Generative AI in Vision

Aug 2024

Cairo, Egypt

- Delivered a talk on Generative AI in Computer Vision for industrial use cases, including fashion modeling, at Dsquares' tech meetup featuring tech leaders and academics.
 - Dell Technologies Hackathon**
Explainable AI in 5G Networks

Feb 2023

Cairo, Egypt

- Presented on explainable AI in 5G applications at Dell's annual hackathon for students and professionals.

Skills

Programming: Python, C/C++, SQL
Software tools: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, Git, Linux

References

Sarah Adel Bargal (Advisor) Assistant Prof., CS Dept., Georgetown University sarah.bargal@georgetown.edu	Marwan Mansour Senior Data Scientist, Dell Technologies Marwan.Mansour@dell.com
Hesham Eraqi Senior Scientist, Amazon heraqi@amazon.com	Cherif Salama Associate Chair, CSE Dept., AUC cherif.salama@aucegypt.edu